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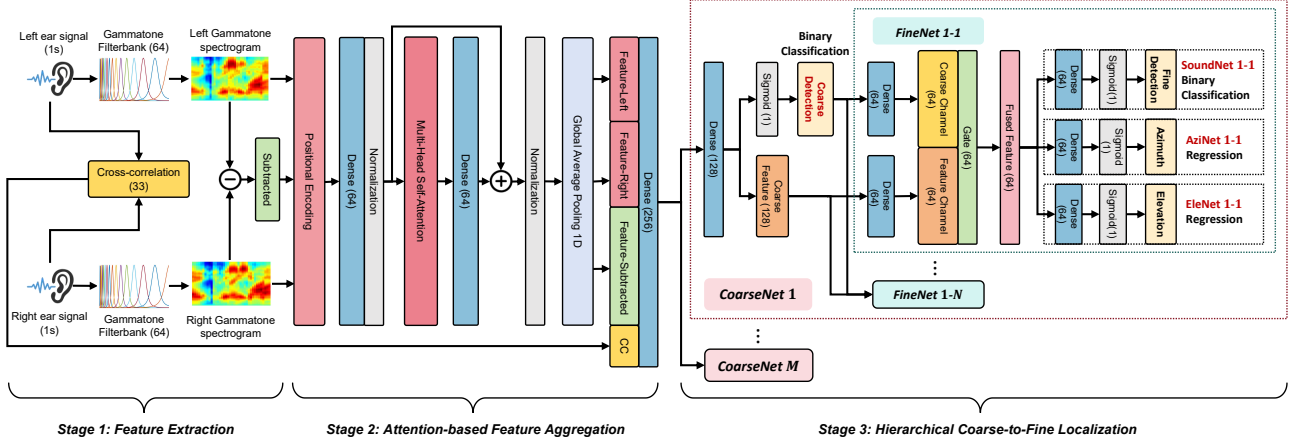


Figure 2: Architecture of proposed AuralNet.

2. Proposed Model

We parameterize sound source DOA on a 3D unit sphere centered at the binaural microphone array, as shown in Fig. 1. Each source is represented by spherical coordinates (φ, θ) , where φ is the azimuth and θ is the elevation. Motivated by human auditory perception characteristics, we design a hybrid spatial encoding strategy. The azimuth is uniformly divided into M sectors (each spanning $360^\circ/M$) to leverage precise binaural time-difference cues. For elevation, we adopt a symmetric partitioning scheme and divide the range $[-75^\circ, 75^\circ]$ into N equal bands (each spanning $150^\circ/N$), avoiding boundaries near $\theta = 0^\circ$. This design specifically addresses the perceptual ambiguity near the horizontal plane ($\theta = 0^\circ$), where human hearing struggles to distinguish small elevation differences due to symmetric pinna filtering effects. As shown in Fig. 1(b) with $M = 8$ and $N = 3$, the azimuth is partitioned into eight 45° sectors, and the elevation into three regions: $[-75^\circ, -25^\circ]$ (low), $[-25^\circ, 25^\circ]$ (central), and $[25^\circ, 75^\circ]$ (high), resulting in 24 sectors in total. This 3D sectorization enables localization of up to $M \times N$ concurrent sound sources (assuming at most one active source per sector).

2.1. AuralNet Overview

Figure 2 illustrates AuralNet. Inspired by the human auditory system, AuralNet integrates a hierarchical coarse-to-fine multi-task learning framework, enabling simultaneous estimation of azimuth and elevation for multiple sound sources. The model is structured around three core stages: Feature Extraction, Attention-based Feature Aggregation, and Hierarchical Coarse-to-Fine Multi-Task Learning.

2.2. Feature Extraction

The first stage of AuralNet transforms binaural signals into time-frequency representations. To emulate the frequency response of the basilar membrane in the cochlea, we apply a Gammatone filterbank [33] with 64 filters that span the frequency range from 50 to 8,000 Hz. The signals are sampled at 16 kHz, and each 1-second signal from the left and right ears is framed into 50 ms segments with 50% overlap using a Hamming window. These segments are passed through the filterbank, producing log-Gammatone power spectrograms for both left and right ear signals. The output tensor shape for each ear is $(T, 64)$, where $T = 39$ represents the number of time frames.

In parallel, the GCC-PHAT algorithm is applied to extract 33 cross-correlation (CC) coefficients from the left- and right-ear signals. The coefficients represent the interaural time differences within a range of ± 1 ms, capturing the spatial relationship between the left and right ears.

2.3. Attention-based Feature Aggregation

Next, we aggregate the features extracted in the previous stage using a self-attention mechanism, which allows the model to adaptively focus on the most important temporal and spectral cues within the features, facilitating more accurate SSL. To be specific, we begin by feeding the left and right Gammatone features, along with the CC features, into a dedicated attention module. In detail, the Gammatone features for each ear are first processed by individual self-attention modules to capture relevant temporal and spectral patterns in each channel. The attention mechanism is enhanced with positional encoding to maintain the temporal structure of the data, which is crucial for sequential information processing. Meanwhile, the differences between the left and right Gammatone features are computed and passed through another attention module to capture interaural disparities, which are discriminative for SSL.

After attention-based processing, global average pooling (GAP) is applied to summarize the temporal information into a global feature vector. This pooling is performed on the outputs of the self-attention modules for the Gammatone features and the interaural feature difference, which are then concatenated with the CC features to form the aggregated feature vector. Finally, the concatenated feature vector is passed through a Multi-Layer Perceptron (MLP), which further refines the aggregated representation before it is sent to the hierarchical coarse-to-fine localization module.

2.4. Hierarchical Coarse-to-Fine Multi-Task Learning

The third stage of our model introduces a hierarchical coarse-to-fine multi-task learning framework, designed to improve SSL by jointly estimating azimuth and elevation. This approach is structured into two levels: a coarse level for preliminary predictions and a fine level for detailed refinement. The model leverages a gating mechanism and a custom weighted loss function to enhance SSL performance.

Coarse Level: At the coarse level, the 360° azimuth range is divided into M coarse sectors, each representing an azimuth division. The model generates a preliminary sound detec-

tion probability for each sector (a binary classification), which guides the subsequent finer-grained localization process.

Fine Level: Each coarse branch is further divided into N fine branches, corresponding to specific elevation ranges (fine sectors). These fine branches work in tandem to refine the coarse azimuth predictions. A *gating mechanism* is designed to combine coarse-level and fine-grained features:

$$\mathbf{g} = \sigma(\mathbf{W}_g \cdot \text{concat}(\mathbf{f}_{\text{coarse}}, \mathbf{f}_{\text{fine}}) + \mathbf{b}_g), \quad (1)$$

where $\mathbf{g}$ is the gate output, σ the sigmoid activation function, $\mathbf{W}_g$ the learnable weight matrix, $\mathbf{b}_g$ the bias term, and $\mathbf{f}_{\text{fine}}$ and $\mathbf{f}_{\text{coarse}}$ represent the fine and coarse features, respectively. The final fused feature for each fine branch is

$$\mathbf{f}_{\text{fused}} = \mathbf{g} \cdot \mathbf{f}_{\text{fine}} + (1 - \mathbf{g}) \cdot \mathbf{f}_{\text{coarse}}. \quad (2)$$

This gating mechanism allows adaptive fusion of coarse and fine features, dynamically adjusting the contribution to ensure the model focuses on the most relevant information.

Each fine branch produces three predictions for its corresponding sector: a binary classification for sound detection (indicating the presence of a source), a normalized azimuth angle, and a normalized elevation angle. The predictions from all N fine branches are concatenated with the coarse detection, forming $3N + 1$ predictions for each coarse branch. Finally, the outputs from all M coarse branches are concatenated along the first dimension to produce the final $(M, 3N + 1)$ predictions, providing 3D SSL results across all $M \times N$ sectors.

Multi-Task Loss Optimization: The network is trained with a multi-task loss function that integrates classification and regression tasks at both coarse and fine levels. For each coarse branch $i \in \{1, 2, \dots, M\}$ and each fine sector $j \in \{1, 2, \dots, N\}$ within branch i , the sound detection loss is computed by using binary cross-entropy (BCE):

$$\mathcal{L}_{\text{det}}^{i,j} = \text{BCE}(\hat{y}_{\text{det}}^{i,j}, y_{\text{det}}^{i,j}), \quad (3)$$

where $\hat{y}_{\text{det}}^{i,j}$ is the predicted detection result, and $y_{\text{det}}^{i,j}$ is the ground-truth label. The azimuth and elevation losses are computed using masked mean absolute error (MAE):

$$\mathcal{L}_{\text{azi}}^{i,j} = \frac{1}{C_{\text{valid}}^{i,j}} \sum_k |\hat{y}_{\text{azi}}^{i,j,k} - y_{\text{azi}}^{i,j,k}| \cdot m_k^{i,j}, \quad (4)$$

$$\mathcal{L}_{\text{ele}}^{i,j} = \frac{1}{C_{\text{valid}}^{i,j}} \sum_k |\hat{y}_{\text{ele}}^{i,j,k} - y_{\text{ele}}^{i,j,k}| \cdot m_k^{i,j}, \quad (5)$$

where $\hat{y}_{\text{azi}}^{i,j,k}$ and $\hat{y}_{\text{ele}}^{i,j,k}$ are the predicted azimuth and elevation for the k -th sample, $y_{\text{azi}}^{i,j,k}$ and $y_{\text{ele}}^{i,j,k}$ the corresponding ground-truth labels. The term $m_k^{i,j}$ is a binary mask indicating the presence of a sound source in the j -th sector of the i -th coarse branch for the k -th sample, and $C_{\text{valid}}^{i,j}$ is the number of valid (active) sectors across all samples. This masking ensures that angular errors are only computed for sectors where a sound source is present, preventing optimization based on irrelevant regions.

To further enhance localization, we add a coarse-level supervision term using BCE loss:

$$\mathcal{L}_{\text{coarse}} = \text{BCE}(\hat{y}_{\text{coarse}}^i, y_{\text{coarse}}^i), \quad (6)$$

where $\hat{y}_{\text{coarse}}^i$ is the predicted coarse-level detection result for the i -th coarse branch, y_{coarse}^i the corresponding ground-truth label. The total multi-task loss is a weighted sum of all losses:

$$\mathcal{L} = \sum_{i=1}^M \left[\delta \mathcal{L}_{\text{coarse}}^i + \sum_{j=1}^N \left(\alpha \mathcal{L}_{\text{det}}^{i,j} + \beta \mathcal{L}_{\text{azi}}^{i,j} + \gamma \mathcal{L}_{\text{ele}}^{i,j} \right) \right], \quad (7)$$

where δ , α , β , and γ are empirically determined weights, set to 0.2, 0.2, 0.5, 0.3 in our experiments. This weighted masked loss function ensures that the network focuses on optimizing localization accuracy only where sound sources are present, leveraging both coarse and fine-level outputs.

3. Experimental Setup

3.1. Dataset

Clean binaural signals were synthesized by convolving anechoic TIMIT speech [34] with head-related transfer functions (HRTFs) from the ITA database [35], using subject MRT05. The DOA space spanned 0° to 360° in azimuth and -65° to 75° in elevation, both at 5° resolution, yielding 2,088 directions. The 3D space was divided into $M = 8$ coarse azimuthal sectors (45° each), each further split into $N = 3$ fine sectors. Elevation coverage varied across low (81), central (90), and high (90) regions. Sources were placed 1 m from the microphone center, with a 0.14 m inter-microphone distance. To simulate reverberant conditions, room impulse responses (RIRs) were generated using the image method [36] and convolved with the clean binaural signals. Diffuse noise from the PCAFETER subset of the DEMAND dataset [37] was added to simulate a noisy, reverberant, cafeteria-like environment at SNRs of 20, 10, and 0 dB.

The training set included three configurations: *1-talker* (25,056 samples), *2-talker* (43,848 samples), and *3-talker* (29,232 samples). Samples were generated following the above procedure, with multi-source cases enforcing a minimum azimuthal separation of 45° . Random room layouts were used, simulating small ($[5, 5, 3]$ m / $[6, 6, 4]$ m, $T_{60} = 0.3/0.4$ s), medium ($[10, 8, 4]$ m / $[12, 10, 5]$ m, $T_{60} = 0.5/0.6$ s), and large rooms ($[15, 10, 5]$ m / $[13, 9, 5]$ m, $T_{60} = 0.7/0.8$ s). Validation used unseen rooms and speech, while testing covered both seen and unseen configurations across all source/SNR conditions. In total, the training set comprised 98,136 samples across the three configurations. The validation and test sets contained 2,088 single-source and 2,800 multi-source samples, respectively. The Adam optimizer (learning rate 0.001, batch size 200) was used, and training was stopped after 100 epochs with early stopping if no improvement was observed after 10 epochs on validation loss.

3.2. Evaluation Metrics

We evaluate AuralNet using three metrics: Detection Accuracy, F1 Score, and Detection-aware Angular Error.

Detection Accuracy measures the binary detection accuracy of the fine branch's sound detection output.

F1 Score is the harmonic mean of precision and recall, measuring the model's ability to detect sound sources while minimizing false positives and negatives.

Detection-aware Angular Error (DAE) evaluates the average angular error for correctly detected sectors, calculated as:

$$\text{DAE} = \frac{1}{N_{\text{valid}}} \sum_{n=1}^{N_{\text{valid}}} (|\hat{\varphi}_n - \varphi_n| + |\hat{\theta}_n - \theta_n|), \quad (8)$$

where $\hat{\varphi}_n$ and $\hat{\theta}_n$ are the predicted azimuth and elevation (denormalized to degrees), and φ_n and θ_n are the ground-truth values. N_{valid} is the number of valid detections, computed as

$$N_{\text{valid}} = \sum_{k=1}^{M \cdot N} \mathbb{I}(\hat{y}_{\text{det}}^k = 1 \text{ and } y_{\text{det}}^k = 1), \quad (9)$$

Table 1: Performance Comparison (Seen / Unseen Rooms)

Source Type		1-talker			2-talker			3-talker			Avg.
SNR (dB)		20	10	0	20	10	0	20	10	0	
Sound Detection Accuracy (%)	DeepEar [26]	98.24 / 98.33	98.35 / 98.34	97.91 / 97.81	93.84 / 93.73	93.87 / 94.03	93.15 / 93.13	89.32 / 89.33	89.33 / 89.21	88.58 / 88.84	93.63
	AuralNet	99.40 / 99.39	99.22 / 99.23	98.79 / 98.71	95.68 / 95.52	95.50 / 95.46	94.56 / 94.51	90.84 / 90.69	90.57 / 90.43	89.55 / 89.65	94.87
	AuralNet-NC	99.30 / 99.26	99.16 / 99.02	98.55 / 98.44	95.36 / 95.20	95.03 / 95.15	94.26 / 94.11	90.47 / 90.49	90.18 / 89.98	89.22 / 89.28	94.58
	AuralNet-NH	99.33 / 99.28	99.13 / 99.07	98.53 / 98.54	95.36 / 95.17	95.03 / 95.05	94.17 / 94.02	90.38 / 90.40	90.14 / 90.12	89.28 / 89.35	94.58
	AuralNet-RL	99.01 / 98.96	98.76 / 98.77	98.19 / 98.05	95.05 / 94.84	94.71 / 94.72	93.71 / 93.76	90.07 / 90.09	89.71 / 89.72	88.91 / 89.08	94.23
Sound Detection F1 score (%)	DeepEar [26]	77.72 / 78.88	78.99 / 78.94	71.72 / 70.70	55.29 / 54.72	54.07 / 55.85	43.85 / 43.69	40.55 / 40.89	38.79 / 37.98	29.42 / 30.66	54.59
	AuralNet	92.88 / 92.71	90.70 / 90.70	85.47 / 84.38	72.68 / 71.73	71.34 / 70.93	63.39 / 62.84	56.64 / 55.90	55.04 / 54.16	46.49 / 46.89	70.27
	AuralNet-NC	91.68 / 91.23	90.11 / 88.35	82.76 / 81.32	70.52 / 69.90	68.30 / 69.30	61.03 / 60.20	54.69 / 54.90	52.50 / 51.56	44.22 / 44.57	68.17
	AuralNet-NH	91.97 / 91.39	89.74 / 88.99	82.54 / 82.61	70.56 / 69.46	68.20 / 68.45	60.60 / 59.66	53.96 / 54.24	52.44 / 52.11	44.31 / 45.12	68.13
	AuralNet-RL	88.07 / 87.49	85.06 / 85.08	77.78 / 76.05	68.03 / 66.66	65.35 / 65.45	56.18 / 56.76	51.07 / 51.42	48.35 / 48.48	40.66 / 42.01	64.44
Azimuth DAE (°)	DeepEar [26]	5.99 / 6.09	5.98 / 6.04	7.37 / 7.46	8.19 / 8.34	8.62 / 8.74	9.29 / 9.14	10.05 / 9.91	10.35 / 9.91	10.22 / 10.22	8.44
	AuralNet	1.16 / 1.26	1.31 / 1.37	1.74 / 1.78	2.81 / 2.74	2.89 / 2.96	3.36 / 3.44	4.05 / 3.93	4.23 / 4.12	4.64 / 4.77	2.92
	AuralNet-NC	1.71 / 1.81	1.83 / 1.92	2.24 / 2.45	3.21 / 3.25	3.32 / 3.56	3.96 / 3.93	4.68 / 4.78	4.77 / 4.78	5.36 / 5.30	3.49
	AuralNet-NH	1.64 / 1.67	1.76 / 1.83	2.17 / 2.23	3.20 / 3.26	3.36 / 3.34	4.01 / 3.96	4.87 / 4.69	4.88 / 4.83	5.36 / 5.37	3.47
	AuralNet-RL	18.00 / 17.95	17.71 / 17.83	18.25 / 18.22	18.50 / 18.72	18.90 / 18.92	18.72 / 18.91	19.96 / 20.08	20.39 / 20.23	19.85 / 19.89	18.95
Elevation DAE (°)	DeepEar [26]	7.33 / 7.20	7.65 / 7.61	9.12 / 9.03	10.06 / 9.86	10.15 / 10.08	10.52 / 10.47	10.88 / 10.83	11.17 / 10.78	11.36 / 11.00	9.73
	AuralNet	1.68 / 1.81	1.80 / 1.94	2.36 / 2.52	3.86 / 3.85	4.04 / 4.07	4.59 / 4.75	5.35 / 5.63	5.67 / 5.53	6.11 / 6.24	3.99
	AuralNet-NC	2.38 / 2.43	2.47 / 2.64	3.12 / 3.20	4.75 / 4.59	4.73 / 4.84	5.23 / 5.55	6.14 / 6.27	6.45 / 6.23	6.75 / 7.02	4.71
	AuralNet-NH	2.15 / 2.21	2.27 / 2.36	2.89 / 3.10	4.42 / 4.52	4.69 / 4.68	5.23 / 5.42	6.11 / 6.22	6.22 / 6.09	6.70 / 7.03	4.57
	AuralNet-RL	19.00 / 18.96	18.92 / 18.74	19.05 / 19.32	19.92 / 20.26	19.52 / 20.23	19.66 / 19.53	20.36 / 20.63	20.21 / 19.71	20.09 / 19.76	19.66

where $M \cdot N$ is the total number of fine sectors, $\hat{y}_{\text{det}}^k$ is the detection result (1 for detection, 0 for no detection), and y_{det}^k is the ground-truth label.

4. Evaluation Results and Discussion

We evaluated AuralNet against the recent sector-based multi-BSSL system, DeepEar [26]. To ensure fair comparison, we extend DeepEar by adapting its 2D sector division into a 24-sector 3D framework and replacing its distance estimation sub-network with our elevation estimator, resulting in a model size of 1.96 M parameters. To assess the effectiveness of main components in AuralNet, we compared the performance of AuralNet (1.11 M parameters) with its three variants:

- **AuralNet-NoCoarse (AuralNet-NC)** removes the coarse detection task, bypassing the coarse output and gating mechanism, resulting in 1.00 M parameters.
- **AuralNet-NonHierarchical (AuralNet-NH)** builds upon AuralNet-NC and adopts a non-hierarchical structure with 24 identical sub-networks for output prediction, increasing the model size to 1.53 M.
- **AuralNet-RegularLoss (AuralNet-RL)** uses the standard loss function, with BCE for both coarse and fine sound detection and MAE for azimuth and elevation prediction, maintaining the same architecture as AuralNet (1.11 M).

The results summarized in Table 1 indicate that AuralNet consistently outperforms DeepEar across all tasks and noisy-reverberant conditions, with significant improvements in F1 score and DAE in both azimuth and elevation estimation. AuralNet achieved the highest detection accuracy and F1 score across all SNRs and room configurations. At 20 dB SNR, AuralNet’s F1 score for 1-talker in unseen rooms was 92.71%, outperforming DeepEar’s 78.88% by nearly 14%. The F1 score was lower than accuracy due to class imbalance, with silent sectors dominating the dataset. Despite this, the significant improvement in F1 score highlights AuralNet’s superior ability to handle complex localization tasks.

In terms of DOA estimation, AuralNet demonstrated a substantial advantage in both azimuth and elevation DAE across all scenarios. AuralNet achieved an average azimuth DAE of 2.92° , a remarkable improvement over DeepEar’s 8.44° . Similarly, AuralNet’s average elevation DAE was 3.99° , significantly better than DeepEar’s 9.73° . These results show that AuralNet’s architecture enables more accurate DOA estimation, demonstrating its potential for real-world BSSL tasks.

The comparison of AuralNet with three variants showed important performance trade-offs. AuralNet-NC, which omitted the coarse detection task, exhibited slightly lower performance, particularly in sound detection and F1 score. This highlights the crucial role of the coarse detection task and its supervisory effect through the coarse loss, which is beneficial for the subsequent fine tasks. AuralNet-NH, despite having a larger model size, showed similar performance to AuralNet-NC in all tasks, underscoring the efficiency of the hierarchical structure. This suggests that the hierarchical approach can reduce model size and redundancy without sacrificing performance. Lastly, AuralNet-RL, which used a standard loss function, performed the worst, especially in DOA estimation. This stark performance drop highlights the effectiveness of AuralNet’s masked loss function in improving DOA accuracy.

Overall, AuralNet demonstrated significant improvements over DeepEar in both sound detection and localization tasks. Its architecture, including the coarse detection task, hierarchical structure, and custom loss function, contributed to its superior performance, making it a promising solution for real-world multi-BSSL scenarios.

5. Conclusions

We have proposed AuralNet, a novel hierarchical attention-based model for 3D BSSL of multiple concurrent speakers. Unlike existing methods limited to 2D or single-source localization, AuralNet jointly estimates azimuth and elevation in 3D, requiring no prior knowledge of the number of sources. By utilizing a coarse-to-fine multi-task learning strategy and integrating self-attention mechanisms for binaural feature aggregation, AuralNet significantly outperforms DeepEar [26], a recent sector-based multi-BSSL system, in noisy-reverberant environments, especially in low-SNR and multi-speaker scenarios. Future work will focus on improving the model’s generalizability to more challenging acoustic environments and to optimize its performance for real-world embodied applications.

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